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|                                                                                                                                                                                                                       |  |                     |  |                       |  |                                    |  |             |  |                        |  |                                   |  |                                    |  |                                    |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------|--|-----------------------|--|------------------------------------|--|-------------|--|------------------------|--|-----------------------------------|--|------------------------------------|--|------------------------------------|--|
| 1                                                                                                                                                                                                                     |  | 2                   |  | 3                     |  | 4                                  |  | 5           |  | 6                      |  | 7                                 |  | 8                                  |  |                                    |  |
| A                                                                                                                                                                                                                     |  |                     |  |                       |  |                                    |  |             |  |                        |  |                                   |  |                                    |  |                                    |  |
| B                                                                                                                                                                                                                     |  |                     |  |                       |  |                                    |  |             |  |                        |  |                                   |  |                                    |  |                                    |  |
| C                                                                                                                                                                                                                     |  |                     |  |                       |  |                                    |  |             |  |                        |  |                                   |  |                                    |  |                                    |  |
| D                                                                                                                                                                                                                     |  |                     |  |                       |  |                                    |  |             |  |                        |  |                                   |  |                                    |  |                                    |  |
| E                                                                                                                                                                                                                     |  |                     |  |                       |  |                                    |  |             |  |                        |  |                                   |  |                                    |  |                                    |  |
| F                                                                                                                                                                                                                     |  |                     |  |                       |  |                                    |  |             |  |                        |  |                                   |  |                                    |  |                                    |  |
| <div>UniSec</div> <div>Medium Voltage Switchgear</div> <div>PT MULTIMAS NABATI ASAHAN</div> <div>MVD-20 &amp; MVD-21 20kV Boiler Plant</div> <div>SCHEMATIC DIAGRAM</div> <div>SDS RISER WITH VT</div> <div>H06</div> |  |                     |  |                       |  |                                    |  |             |  |                        |  |                                   |  |                                    |  |                                    |  |
|                                                                                                                                                                                                                       |  |                     |  | Date Prep. 15.11.2018 |  | PROJECT TITLE                      |  |             |  | ABB EPMV               |  | COVER SHEET                       |  | Scale                              |  |                                    |  |
|                                                                                                                                                                                                                       |  |                     |  | Prepared YR           |  | MV Switchgear for MNA Serang Plant |  |             |  | PT. ABB Sakti Industri |  | MVD-20 & MVD-21 20kV Boiler Plant |  | Customer PT Multimas Nabati Asahan |  |                                    |  |
| 00                                                                                                                                                                                                                    |  | Issued for Approval |  | 11.01.19 YR           |  |                                    |  |             |  |                        |  | Checked AW                        |  | SCHEMATIC DIAGRAM                  |  | Drawing Nbr. IDUNISEC-BE318042-301 |  |
| Ind.                                                                                                                                                                                                                  |  | Revision            |  | Date                  |  | Name                               |  | Approved AI |  | Based on: BE318042     |  |                                   |  | Customer Doc. Nbr.                 |  | Cont. 16                           |  |
| 1                                                                                                                                                                                                                     |  | 2                   |  | 3                     |  | 4                                  |  | 5           |  | 6                      |  | 7                                 |  | 8                                  |  |                                    |  |

UniSec

Medium Voltage Switchgear  
PT MULTIMAS NABATI ASAHAN  
MVD-20 & MVD-21 20kV Boiler Plant  
SCHEMATIC DIAGRAM  
SDS RISER WITH VT  
H06

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|   | 1                          | 2                   | 3                                          | 4    | 5          | 6          | 7                                  | 8 |                        |      |                                   |  |                    |                           |       |    |
| A | SHEET INDEX                |                     |                                            |      |            |            |                                    |   | A                      |      |                                   |  |                    |                           |       |    |
| B | PAGE                       |                     | PAGE NAME                                  |      |            | REVISION   |                                    |   | B                      |      |                                   |  |                    |                           |       |    |
|   | ==MVD-20_MVD-21=H06&310/1  |                     | COVER SHEET                                |      |            | 00         |                                    |   |                        |      |                                   |  |                    |                           |       |    |
|   | ==MVD-20_MVD-21=H06&310/2  |                     | Index of Sheet                             |      |            | 00         |                                    |   |                        |      |                                   |  |                    |                           |       |    |
|   | ==MVD-20_MVD-21=H06&310/3  |                     | REFERENCE DESIGNATIONS                     |      |            | 00         |                                    |   |                        |      |                                   |  |                    |                           |       |    |
|   | ==MVD-20_MVD-21=H06&310/4  |                     | REFERENCE DESIGNATIONS                     |      |            | 00         |                                    |   |                        |      |                                   |  |                    |                           |       |    |
|   | ==MVD-20_MVD-21=H06&310/5  |                     | REFERENCE DESIGNATIONS                     |      |            | 00         |                                    |   |                        |      |                                   |  |                    |                           |       |    |
|   | ==MVD-20_MVD-21=H06&310/6  |                     | REFERENCE DESIGNATIONS                     |      |            | 00         |                                    |   |                        |      |                                   |  |                    |                           |       |    |
|   | ==MVD-20_MVD-21=H06&310/7  |                     | DISTRIBUTION OF AUXILIARY CIRCUITS         |      |            | 00         |                                    |   |                        |      |                                   |  |                    |                           |       |    |
|   | ==MVD-20_MVD-21=H06&310/8  |                     | MAIN CIRCUITS                              |      |            | 00         |                                    |   |                        |      |                                   |  |                    |                           |       |    |
|   | ==MVD-20_MVD-21=H06&310/9  |                     | LV DOOR LAYOUT                             |      |            | 00         |                                    |   |                        |      |                                   |  |                    |                           |       |    |
|   | ==MVD-20_MVD-21=H06&310/10 |                     | LV BOX LAYOUT                              |      |            | 00         |                                    |   |                        |      |                                   |  |                    |                           |       |    |
|   | ==MVD-20_MVD-21=H06&310/11 |                     | PART LIST                                  |      |            | 00         |                                    |   |                        |      |                                   |  |                    |                           |       |    |
|   | ==MVD-20_MVD-21=H06&310/12 |                     | Terminal diagram ==MVD-20_MVD-21=H06+-XDI  |      |            |            |                                    |   |                        |      |                                   |  |                    |                           |       |    |
|   | ==MVD-20_MVD-21=H06&310/13 |                     | Terminal diagram ==MVD-20_MVD-21=H06+-XDI2 |      |            |            |                                    |   |                        |      |                                   |  |                    |                           |       |    |
|   | ==MVD-20_MVD-21=H06&310/14 |                     | Terminal diagram ==MVD-20_MVD-21=H06+-XDI6 |      |            |            |                                    |   |                        |      |                                   |  |                    |                           |       |    |
|   | ==MVD-20_MVD-21=H06&310/15 |                     | Terminal diagram ==MVD-20_MVD-21=H06+-XDV  |      |            |            |                                    |   |                        |      |                                   |  |                    |                           |       |    |
| C | ==MVD-20_MVD-21=H06&310/16 |                     | Terminal diagram ==MVD-20_MVD-21=H06+-XDX  |      |            |            |                                    |   | C                      |      |                                   |  |                    |                           |       |    |
| D |                            |                     |                                            |      |            |            |                                    |   | D                      |      |                                   |  |                    |                           |       |    |
| E |                            |                     |                                            |      |            |            |                                    |   | E                      |      |                                   |  |                    |                           |       |    |
| F |                            |                     |                                            |      | Date Prep. | 15.11.2018 | PROJECT TITLE                      |   | ABB                    | EPMV | Index of Sheet                    |  | Scale              |                           | = H06 | F  |
|   |                            |                     |                                            |      | Prepared   | YR         | MV Switchgear for MNA Serang Plant |   |                        |      | MVD-20 & MVD-21 20kV Boiler Plant |  | Customer           | PT Multimas Nabati Asahan | & 310 |    |
|   | 00                         | Issued for Approval | 11.01.19                                   | YR   | Checked    | AW         |                                    |   | PT. ABB Sakti Industri |      | SCHEMATIC DIAGRAM                 |  | Drawing Nbr.       | IDUNISEC-BE318042-301     | Sh.   | 2  |
|   | Ind.                       | Revision            | Date                                       | Name | Approved   | AI         | Based on: BE318042                 |   |                        |      | SDS RISER WITH VT                 |  | Customer Doc. Nbr. |                           | Cont. | 16 |
|   | 1                          | 2                   | 3                                          | 4    | 5          | 6          | 7                                  | 8 |                        |      |                                   |  |                    |                           |       |    |

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|---|---------------------------------------------------------------------------------|--|--|----|-----------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|---|
| A | REFERENCE DESIGNATION OF OBJECTS IN ELECTRICAL DOCUMENTS                        |  |  |    |                                                                                                                                                     |  |  |  |  |  |  |  |  |  | A |
|   | (IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001) |  |  |    |                                                                                                                                                     |  |  |  |  |  |  |  |  |  |   |
| B | DESIGNATION                                                                     |  |  |    | DESCRIPTION                                                                                                                                         |  |  |  |  |  |  |  |  |  | B |
|   | -AA                                                                             |  |  |    | MULTIFUNCTION UNIT (CENTRAL UNIT)                                                                                                                   |  |  |  |  |  |  |  |  |  |   |
|   | -BAD1,-BAD4                                                                     |  |  |    | CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1                                                                                                      |  |  |  |  |  |  |  |  |  |   |
|   | -BAD2,-BAD5                                                                     |  |  |    | CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2                                                                                                      |  |  |  |  |  |  |  |  |  |   |
|   | -BAD3,-BAD6                                                                     |  |  |    | CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3                                                                                                      |  |  |  |  |  |  |  |  |  |   |
|   | -BAR                                                                            |  |  |    | VOLTAGE PROTECTION RELAY                                                                                                                            |  |  |  |  |  |  |  |  |  |   |
|   | -BAS1                                                                           |  |  |    | VOLTAGE SENSOR LOCATED ON PHASE L1                                                                                                                  |  |  |  |  |  |  |  |  |  |   |
|   | -BAS2                                                                           |  |  |    | VOLTAGE SENSOR LOCATED ON PHASE L2                                                                                                                  |  |  |  |  |  |  |  |  |  |   |
|   | -BAS3                                                                           |  |  |    | VOLTAGE SENSOR LOCATED ON PHASE L3                                                                                                                  |  |  |  |  |  |  |  |  |  |   |
|   | -BAT1                                                                           |  |  |    | VOLTAGE TRANSFORMER LOCATED ON PHASE L1                                                                                                             |  |  |  |  |  |  |  |  |  |   |
| C | -BAT2                                                                           |  |  |    | VOLTAGE TRANSFORMER LOCATED ON PHASE L2                                                                                                             |  |  |  |  |  |  |  |  |  | C |
|   | -BAT3                                                                           |  |  |    | VOLTAGE TRANSFORMER LOCATED ON PHASE L3                                                                                                             |  |  |  |  |  |  |  |  |  |   |
|   | -BAT4                                                                           |  |  |    | VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L1                                                                                       |  |  |  |  |  |  |  |  |  |   |
|   | -BAT5                                                                           |  |  |    | VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L2                                                                                       |  |  |  |  |  |  |  |  |  |   |
|   | -BAT6                                                                           |  |  |    | VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L3                                                                                       |  |  |  |  |  |  |  |  |  |   |
|   | -BCD                                                                            |  |  |    | DIFFERENTIAL PROTECTIVE RELAY                                                                                                                       |  |  |  |  |  |  |  |  |  |   |
|   | -BCF                                                                            |  |  |    | FEEDER PROTECTION RELAY                                                                                                                             |  |  |  |  |  |  |  |  |  |   |
|   | -BCG                                                                            |  |  |    | GENERATOR PROTECTION RELAY                                                                                                                          |  |  |  |  |  |  |  |  |  |   |
|   | -BCM                                                                            |  |  |    | MOTOR PROTECTION RELAY                                                                                                                              |  |  |  |  |  |  |  |  |  |   |
|   | -BCN                                                                            |  |  |    | NEUTRAL (RESIDUAL) CURRENT TRANSFORMER                                                                                                              |  |  |  |  |  |  |  |  |  |   |
| D | -BCP                                                                            |  |  |    | TRANSFORMER PROTECTION RELAY                                                                                                                        |  |  |  |  |  |  |  |  |  | D |
|   | -BCR                                                                            |  |  |    | CURRENT PROTECTION RELAY                                                                                                                            |  |  |  |  |  |  |  |  |  |   |
|   | -BCS1                                                                           |  |  |    | CURRENT SENSOR LOCATED ON PHASE L1                                                                                                                  |  |  |  |  |  |  |  |  |  |   |
|   | -BCS2                                                                           |  |  |    | CURRENT SENSOR LOCATED ON PHASE L2                                                                                                                  |  |  |  |  |  |  |  |  |  |   |
|   | -BCS3                                                                           |  |  |    | CURRENT SENSOR LOCATED ON PHASE L3                                                                                                                  |  |  |  |  |  |  |  |  |  |   |
|   | -BCT1                                                                           |  |  |    | CURRENT TRANSFORMER LOCATED ON PHASE L1                                                                                                             |  |  |  |  |  |  |  |  |  |   |
|   | -BCT2                                                                           |  |  |    | CURRENT TRANSFORMER LOCATED ON PHASE L2                                                                                                             |  |  |  |  |  |  |  |  |  |   |
|   | -BCT3                                                                           |  |  |    | CURRENT TRANSFORMER LOCATED ON PHASE L3                                                                                                             |  |  |  |  |  |  |  |  |  |   |
|   | -BCT4,-BCT7                                                                     |  |  |    | CURRENT TRANSFORMER LOCATED ON PHASE L1                                                                                                             |  |  |  |  |  |  |  |  |  |   |
|   | -BCT5,-BCT8                                                                     |  |  |    | CURRENT TRANSFORMER LOCATED ON PHASE L2                                                                                                             |  |  |  |  |  |  |  |  |  |   |
| E | -BCT6,-BCT9                                                                     |  |  |    | CURRENT TRANSFORMER LOCATED ON PHASE L3                                                                                                             |  |  |  |  |  |  |  |  |  | E |
|   | -BCZ                                                                            |  |  |    | DISTANCE PROTECTION RELAY                                                                                                                           |  |  |  |  |  |  |  |  |  |   |
|   | -BEF                                                                            |  |  |    | FREQUENCY PROTECTION RELAY                                                                                                                          |  |  |  |  |  |  |  |  |  |   |
|   | -BER                                                                            |  |  |    | SUPERVISION RELAYS                                                                                                                                  |  |  |  |  |  |  |  |  |  |   |
|   | -BES                                                                            |  |  |    | SYNCHRONIZING RELAY                                                                                                                                 |  |  |  |  |  |  |  |  |  |   |
|   | -BET                                                                            |  |  |    | THERMAL PROTECTION RELAY                                                                                                                            |  |  |  |  |  |  |  |  |  |   |
|   | -BGA1                                                                           |  |  |    | POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT                                                                     |  |  |  |  |  |  |  |  |  |   |
|   | -BGA2                                                                           |  |  |    | POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM) |  |  |  |  |  |  |  |  |  |   |
|   | -BGA3                                                                           |  |  |    | POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT                                                                               |  |  |  |  |  |  |  |  |  |   |
|   | -BGA4                                                                           |  |  |    | POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)                                                  |  |  |  |  |  |  |  |  |  |   |
| F | -BGB1...-BGB3                                                                   |  |  |    | POSITION SWITCHES OF CIRCUIT-BREAKER                                                                                                                |  |  |  |  |  |  |  |  |  | F |
|   | -BGB4                                                                           |  |  |    | POSITION SWITCH OF CIRCUIT-BREAKER: PASSING CONTACT CLOSING MOMENTARILY WHEN THE BREAKER OPENS                                                      |  |  |  |  |  |  |  |  |  |   |
|   |                                                                                 |  |  |    |                                                                                                                                                     |  |  |  |  |  |  |  |  |  |   |
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| A    | -BGI31        |                     | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION                                                     |          |  |      |  |          | -CA            | CAPACITORS                                                                                 |  |           |  |          |  | A |
|      | -BGI32        |                     | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION                                                   |          |  |      |  |          | -CA1           | CAPACITOR FOR FRONT VENTILATOR                                                             |  |           |  |          |  |   |
|      | -BGI33        |                     | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED) |          |  |      |  |          | -CA2           | CAPACITOR FOR REAR VENTILATOR                                                              |  |           |  |          |  |   |
|      | -BGI41...     |                     | POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION                                      |          |  |      |  |          | -EA1           | LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.                                      |  |           |  |          |  |   |
| B    | -BGI42        |                     | POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION                                               |          |  |      |  |          | -EA2           | LIGHTING LAMP LOCATED IN CABLE COMPARTMENT                                                 |  |           |  |          |  | B |
|      | -BGI43...     |                     | POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION                                       |          |  |      |  |          | -EA3           | LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE                                     |  |           |  |          |  |   |
|      | -BGI51        |                     | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION                                            |          |  |      |  |          | -EA4           | LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT                                      |  |           |  |          |  |   |
|      | -BGI52        |                     | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION                                                     |          |  |      |  |          | -EB1           | HEATER LOCATED IN CABLE COMPARTMENT                                                        |  |           |  |          |  |   |
| C    | -BGI53        |                     | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION                                             |          |  |      |  |          | -EB2           | HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT                                              |  |           |  |          |  | C |
|      | -BGK          |                     | POSITION SWITCH OF KEY LOCK                                                                                          |          |  |      |  |          | -EB3           | HEATER LOCATED IN MOTOR CABINET                                                            |  |           |  |          |  |   |
|      | -BGL1         |                     | POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1                                                                      |          |  |      |  |          | -EB4           | HEATER FOR VENTILATION TEST                                                                |  |           |  |          |  |   |
|      | -BGS1...-BGS4 |                     | POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS                                                                         |          |  |      |  |          | -EB5           | HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.                                              |  |           |  |          |  |   |
| D    | -BGS5         |                     | PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION                                                              |          |  |      |  |          | -EB7           | HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE                          |  |           |  |          |  | D |
|      | -BGT1         |                     | POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION                                                      |          |  |      |  |          | -EB8           | HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE                         |  |           |  |          |  |   |
|      | -BGT2         |                     | POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION                                                         |          |  |      |  |          | -FA1           | SURGE ARRESTER LOCATED ON PHASE L1                                                         |  |           |  |          |  |   |
|      | -BGT3         |                     | POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION                                           |          |  |      |  |          | -FA2           | SURGE ARRESTER LOCATED ON PHASE L2                                                         |  |           |  |          |  |   |
| E    | -BGT4         |                     | POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION                                                 |          |  |      |  |          | -FA3           | SURGE ARRESTER LOCATED ON PHASE L3                                                         |  |           |  |          |  | E |
|      | -BGT5         |                     | POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION                                                    |          |  |      |  |          | -FCD           | FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS                                    |  |           |  |          |  |   |
|      | -BGT6         |                     | POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION                                                          |          |  |      |  |          | -FCF1          | MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1                                                    |  |           |  |          |  |   |
|      | -BGT7         |                     | POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION                                                             |          |  |      |  |          | -FCF2          | MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2                                                    |  |           |  |          |  |   |
| F    | -BGT8         |                     | POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION                                                      |          |  |      |  |          | -FCF3          | MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3                                                    |  |           |  |          |  | F |
|      | -BGT11        |                     | PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION                                                                |          |  |      |  |          | -FCM1          | MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER |  |           |  |          |  |   |
|      | -BGT12        |                     | PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION                                                                   |          |  |      |  |          | -FCM2...-FCM10 | MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS                            |  |           |  |          |  |   |
|      | -BGV1         |                     | POSITION SWITCH OF FRONT VENTILATOR                                                                                  |          |  |      |  |          | -FCM11         | MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1                                |  |           |  |          |  |   |
| G    | -BGV2         |                     | POSITION SWITCH OF REAR VENTILATOR                                                                                   |          |  |      |  |          | -FCM12         | MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2                                |  |           |  |          |  | G |
|      | -BM           |                     | HYGROSTAT                                                                                                            |          |  |      |  |          | -FCM13         | MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3                                |  |           |  |          |  |   |
|      | -BPA4         |                     | PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING                                        |          |  |      |  |          | -FCT           | THERMAL OVERLOAD RELEASES                                                                  |  |           |  |          |  |   |
|      | -BPA5         |                     | PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING               |          |  |      |  |          | -GA            | GENERATORS                                                                                 |  |           |  |          |  |   |
| H    | -BPA6         |                     | PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING      |          |  |      |  |          | -GQ1           | FRONT VENTILATOR                                                                           |  |           |  |          |  | H |
|      | -BPA7         |                     | PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING                |          |  |      |  |          | -GQ2           | REAR VENTILATOR                                                                            |  |           |  |          |  |   |
|      | -BPS1         |                     | PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER                                 |          |  |      |  |          | -GQ3           | VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK                                                |  |           |  |          |  |   |
|      | -BPS2         |                     | PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2                                                          |          |  |      |  |          | -KFA1          | AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE                                                |  |           |  |          |  |   |
| I    | -BPS3         |                     | PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3                                                          |          |  |      |  |          | -KFA2          | AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE                                       |  |           |  |          |  | I |
|      | -BR           |                     | FLAME DETECTORS, SMOKE DETECTORS                                                                                     |          |  |      |  |          | -KFA3...-KFA9  | AUXILIARY RELAYS OR CONTACTORS                                                             |  |           |  |          |  |   |
|      | -BT           |                     | THERMOSTAT                                                                                                           |          |  |      |  |          | -KFC           | CLOSING RELAYS OR CONTACTORS                                                               |  |           |  |          |  |   |
|      | -BT1          |                     | THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS                                                                        |          |  |      |  |          | -KFI           | INTEGRATED CIRCUITS                                                                        |  |           |  |          |  |   |
| J    | -BT2          |                     | THERMOSTAT FOR ENERGIZATION OF VENTILATORS                                                                           |          |  |      |  |          | -KFL           | LOCKOUT RELAY                                                                              |  |           |  |          |  | J |
|      | -BT3          |                     | THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM                                                                         |          |  |      |  |          | -KFM           | MICROPROCESSORS                                                                            |  |           |  |          |  |   |
|      | -BU           |                     | BUCHHOLZ RELAY                                                                                                       |          |  |      |  |          | -KFN           | ANTIPUMPING RELAYS                                                                         |  |           |  |          |  |   |
|      | -BX1          |                     | UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING                                                                   |          |  |      |  |          | -KFO           | OPENING RELAYS OR CONTACTORS                                                               |  |           |  |          |  |   |
| K    | -BX2          |                     | ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING                                                     |          |  |      |  |          | -KFP           | PROGRAMMABLE LOGIC CONTROLLERS (PLC)                                                       |  |           |  |          |  | K |
|      | -BZ,-BZ4      |                     | MULTIFUNCTION SENSORS                                                                                                |          |  |      |  |          | -KFR           | RECLOSING RELAYS                                                                           |  |           |  |          |  |   |
|      | -BZ1          |                     | COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1                                                             |          |  |      |  |          | -KFS           | SYNCHRONIZING DEVICES                                                                      |  |           |  |          |  |   |
|      | -BZ2          |                     | COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2                                                             |          |  |      |  |          | -KFT           | AUXILIARY TIME RELAYS, DELAY ELEMENTS                                                      |  |           |  |          |  |   |
| L    | -BZ3          |                     | COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3                                                             |          |  |      |  |          | -KFZ           | CONTROLLERS                                                                                |  |           |  |          |  | L |
|      |               |                     |                                                                                                                      |          |  |      |  |          | -MBC           | CLOSING RELEASE OF CIRCUIT-BREAKER                                                         |  |           |  |          |  |   |
|      |               |                     |                                                                                                                      |          |  |      |  |          | -MAD           | MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD                                        |  |           |  |          |  |   |
|      |               |                     |                                                                                                                      |          |  |      |  |          | -MAD1          | MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1                                        |  |           |  |          |  |   |
| M    |               |                     |                                                                                                                      |          |  |      |  |          | -MAD2          | MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2                                        |  |           |  |          |  | M |
|      |               |                     |                                                                                                                      |          |  |      |  |          | -MAD9          | MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS                                 |  |           |  |          |  |   |
|      |               |                     |                                                                                                                      |          |  |      |  |          |                |                                                                                            |  |           |  |          |  |   |
|      |               |                     |                                                                                                                      |          |  |      |  |          |                |                                                                                            |  |           |  |          |  |   |
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|      |               |                     |                                                                                                                      |          |  |      |  |          |                |                                                                                            |  |           |  |          |  |   |
| 00   |               | Issued for Approval |                                                                                                                      | 11.01.19 |  | YR   |  | Checked  |                | AW                                                                                         |  |           |  |          |  |   |
| Ind. |               | Revision            |                                                                                                                      | Date     |  | Name |  | Approved |                | AI                                                                                         |  | Based on: |  | BE318042 |  |   |
| 1    |               | 2                   |                                                                                                                      | 3        |  | 4    |  | 5        |                | 6                                                                                          |  | 7         |  | 8        |  |   |

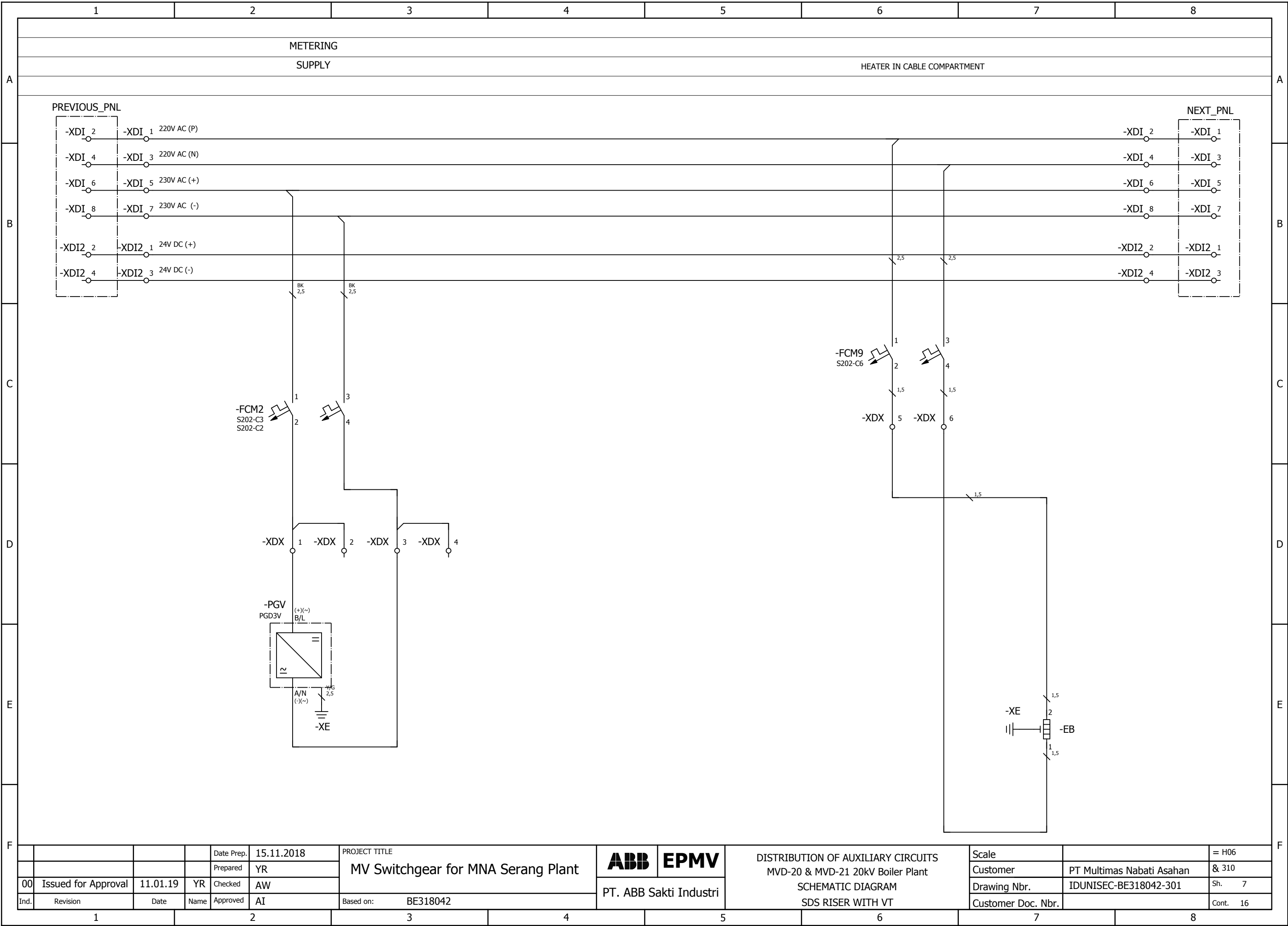
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|---|-------|----------------------------------------------------------------------------------------------------------------|----------|------|------------|------------|-----------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------|--|--------------------|---------------------------|----------|---|
| A | -MAE  | MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE                                                         |          |      |            |            |                                                     | -QBH     | MANUAL CIRCUIT-BREAKERS                                                                                      |      |                                                             |  |                    |                           | A        |   |
|   | -MAE1 | MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2                                               |          |      |            |            |                                                     | -QBL     | LINKS                                                                                                        |      |                                                             |  |                    |                           |          |   |
|   | -MAE8 | MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS         |          |      |            |            |                                                     | -QBM     | MINIATURE SWITCH-DISCONNECTORS                                                                               |      |                                                             |  |                    |                           |          |   |
|   | -MAS  | MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING                                                                     |          |      |            |            |                                                     | -QBS     | SWITCH-DISCONNECTORS                                                                                         |      |                                                             |  |                    |                           |          |   |
|   | -MAT  | MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT                                                         |          |      |            |            |                                                     | -QBT     | TRUCK                                                                                                        |      |                                                             |  |                    |                           |          |   |
|   | -MBC  | CLOSING RELEASE OF CIRCUIT-BREAKER                                                                             |          |      |            |            |                                                     | -QCA     | EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING                                                           |      |                                                             |  |                    |                           |          |   |
|   | -MBC4 | CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS                                                                    |          |      |            |            |                                                     | -QCE     | EARTHING SWITCH                                                                                              |      |                                                             |  |                    |                           |          |   |
|   | -MBO1 | FIRST OPENING RELEASE OF CIRCUIT-BREAKER                                                                       |          |      |            |            |                                                     | -QCE1    | EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)                 |      |                                                             |  |                    |                           |          |   |
|   | -MBO2 | SECOND OPENING RELEASE OF CIRCUIT-BREAKER                                                                      |          |      |            |            |                                                     | -QCE2    | EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)                                                       |      |                                                             |  |                    |                           |          |   |
|   | -MBO3 | OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER |          |      |            |            |                                                     | -QQ      | CLUTCHES                                                                                                     |      |                                                             |  |                    |                           |          |   |
| B | -MBO4 | OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS                                                                    |          |      |            |            |                                                     | -RAA     | ANTIFERRORESONANCE RESISTOR                                                                                  |      |                                                             |  |                    |                           | B        |   |
|   | -MBU  | UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER                                                                        |          |      |            |            |                                                     | -RAD     | DIODES                                                                                                       |      |                                                             |  |                    |                           |          |   |
|   | -PFB  | BLUE SIGNAL LAMPS                                                                                              |          |      |            |            |                                                     | -RAI     | INDUCTORS                                                                                                    |      |                                                             |  |                    |                           |          |   |
|   | -PFF  | FLAG RELAYS                                                                                                    |          |      |            |            |                                                     | -RAR     | RESISTORS                                                                                                    |      |                                                             |  |                    |                           |          |   |
|   | -PFG  | GREEN SIGNAL LAMPS                                                                                             |          |      |            |            |                                                     | -RF      | FILTERS                                                                                                      |      |                                                             |  |                    |                           |          |   |
|   | -PFR  | RED SIGNAL LAMPS                                                                                               |          |      |            |            |                                                     | -RLE1    | ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING                                                    |      |                                                             |  |                    |                           |          |   |
|   | -PFS  | SHORT CIRCUIT INDICATORS                                                                                       |          |      |            |            |                                                     | -RLE2    | ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT                                                       |      |                                                             |  |                    |                           |          |   |
|   | -PFV  | VOLTAGE INDICATORS                                                                                             |          |      |            |            |                                                     | -RLE3    | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE           |      |                                                             |  |                    |                           |          |   |
|   | -PFV1 | VOLTAGE INDICATORS ON FEEDER SIDE                                                                              |          |      |            |            |                                                     | -RLE4    | ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION                                                 |      |                                                             |  |                    |                           |          |   |
|   | -PFV2 | VOLTAGE INDICATORS ON BUSBAR SIDE                                                                              |          |      |            |            |                                                     | -RLE5    | ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD                                             |      |                                                             |  |                    |                           |          |   |
| C | -PFW  | WHITE SIGNAL LAMPS                                                                                             |          |      |            |            |                                                     | -RLE6    | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1             |      |                                                             |  |                    |                           | C        |   |
|   | -PFX  | CROSS INDICATORS, ELECTROMECHANICAL INDICATORS                                                                 |          |      |            |            |                                                     | -RLE7    | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2             |      |                                                             |  |                    |                           |          |   |
|   | -PFY  | YELLOW SIGNAL LAMPS                                                                                            |          |      |            |            |                                                     | -RLE8    | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2 |      |                                                             |  |                    |                           |          |   |
|   | -PGA  | AMMETERS                                                                                                       |          |      |            |            |                                                     | -RLE9    | ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS                                      |      |                                                             |  |                    |                           |          |   |
|   | -PGC  | COUNTERS                                                                                                       |          |      |            |            |                                                     | -RLE11   | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4                     |      |                                                             |  |                    |                           |          |   |
|   | -PGF  | FREQUENCYMETERS                                                                                                |          |      |            |            |                                                     | -RLE12   | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5                     |      |                                                             |  |                    |                           |          |   |
|   | -PGH  | HOURMETERS                                                                                                     |          |      |            |            |                                                     | -SFA     | AMMETRIC SWITCHES                                                                                            |      |                                                             |  |                    |                           |          |   |
|   | -PGI  | PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE                                                           |          |      |            |            |                                                     | -SFC     | CONTROL SWITCHES, CLOSING PUSH-BUTTONS                                                                       |      |                                                             |  |                    |                           |          |   |
|   | -PGI1 | HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)                          |          |      |            |            |                                                     | -SFL     | LOCKING CONTACTS                                                                                             |      |                                                             |  |                    |                           |          |   |
|   | -PGI2 | HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)                              |          |      |            |            |                                                     | -SFO     | OPENING PUSH-BUTTONS                                                                                         |      |                                                             |  |                    |                           |          |   |
| D | -PGJ  | ACTIVE ENERGY METERS                                                                                           |          |      |            |            |                                                     | -SFR     | RESET PUSH-BUTTONS                                                                                           |      |                                                             |  |                    |                           | D        |   |
|   | -PGK  | REACTIVE ENERGY METERS                                                                                         |          |      |            |            |                                                     | -SFS     | SELECTOR SWITCHES                                                                                            |      |                                                             |  |                    |                           |          |   |
|   | -PGM  | MULTIFUNCTION INDICATORS                                                                                       |          |      |            |            |                                                     | -SFT     | TEST PUSH-BUTTONS                                                                                            |      |                                                             |  |                    |                           |          |   |
|   | -PGP  | POWER-FACTOR METERS                                                                                            |          |      |            |            |                                                     | -SFU     | UNLOCKING PUSH-BUTTONS                                                                                       |      |                                                             |  |                    |                           |          |   |
|   | -PGQ  | VARMETERS                                                                                                      |          |      |            |            |                                                     | -SFV     | VOLTMETRIC SWITCHES                                                                                          |      |                                                             |  |                    |                           |          |   |
|   | -PGS  | SYNCHRONOSCOPES                                                                                                |          |      |            |            |                                                     | -TA      | POWER TRANSFORMERS                                                                                           |      |                                                             |  |                    |                           |          |   |
|   | -PGV  | VOLTMETERS                                                                                                     |          |      |            |            |                                                     | -TB      | CONVERTER                                                                                                    |      |                                                             |  |                    |                           |          |   |
|   | -PGW  | WATTMETERS                                                                                                     |          |      |            |            |                                                     | -TB1     | RECTIFIER FOR FIRST OPENING RELEASE                                                                          |      |                                                             |  |                    |                           |          |   |
|   | -PJ   | ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)                                                                      |          |      |            |            |                                                     | -TB2     | RECTIFIER FOR SECOND OPENING RELEASE                                                                         |      |                                                             |  |                    |                           |          |   |
|   | -QAB  | CIRCUIT-BREAKERS                                                                                               |          |      |            |            |                                                     | -TB3     | RECTIFIER FOR CLOSING RELEASE                                                                                |      |                                                             |  |                    |                           |          |   |
| E | -QAC  | CONTACTORS (FOR POWER)                                                                                         |          |      |            |            |                                                     | -TB4     | RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING                                      |      |                                                             |  |                    |                           | E        |   |
|   | -QBD  | DISCONNECTORS                                                                                                  |          |      |            |            |                                                     | -TB5     | RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT                                         |      |                                                             |  |                    |                           |          |   |
|   | -QBD1 | FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)                                                              |          |      |            |            |                                                     | -TB6     | RECTIFIER FOR UNDERVOLTAGE RELEASE                                                                           |      |                                                             |  |                    |                           |          |   |
|   | -QBD2 | REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)                                                               |          |      |            |            |                                                     | -TB7     | ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER                                               |      |                                                             |  |                    |                           |          |   |
|   | -QBD3 | FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)                                                                    |          |      |            |            |                                                     |          |                                                                                                              |      |                                                             |  |                    |                           |          |   |
|   | -QBD4 | TWO-WAY DISCONNECTOR                                                                                           |          |      |            |            |                                                     |          |                                                                                                              |      |                                                             |  |                    |                           |          |   |
|   | -QBD5 | TWO-WAY DISCONNECTOR ON BUSBAR RISER                                                                           |          |      |            |            |                                                     |          |                                                                                                              |      |                                                             |  |                    |                           |          |   |
|   |       |                                                                                                                |          |      |            |            |                                                     |          |                                                                                                              |      |                                                             |  |                    |                           |          |   |
|   |       |                                                                                                                |          |      |            |            |                                                     |          |                                                                                                              |      |                                                             |  |                    |                           |          |   |
|   |       |                                                                                                                |          |      |            |            |                                                     |          |                                                                                                              |      |                                                             |  |                    |                           |          |   |
| F |       |                                                                                                                |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br>MV Switchgear for MNA Serang Plant |          | ABB                                                                                                          | EPMV | REFERENCE DESIGNATIONS<br>MVD-20 & MVD-21 20kV Boiler Plant |  | Scale              |                           | = H06    | F |
|   |       |                                                                                                                |          |      | Prepared   | YR         |                                                     |          |                                                                                                              |      |                                                             |  | Customer           | PT Multimas Nabati Asahan | & 310    |   |
|   | 00    | Issued for Approval                                                                                            | 11.01.19 | YR   | Checked    | AW         |                                                     |          | PT. ABB Sakti Industri                                                                                       |      | SCHEMATIC DIAGRAM<br>SDS RISER WITH VT                      |  | Drawing Nbr.       | IDUNISEC-BE318042-301     | Sh. 5    |   |
|   | Ind.  | Revision                                                                                                       | Date     | Name | Approved   | AI         | Based on:                                           | BE318042 |                                                                                                              |      |                                                             |  | Customer Doc. Nbr. |                           | Cont. 16 |   |
|   | 1     |                                                                                                                | 2        |      | 3          |            | 4                                                   |          | 5                                                                                                            |      | 6                                                           |  | 7                  |                           | 8        |   |

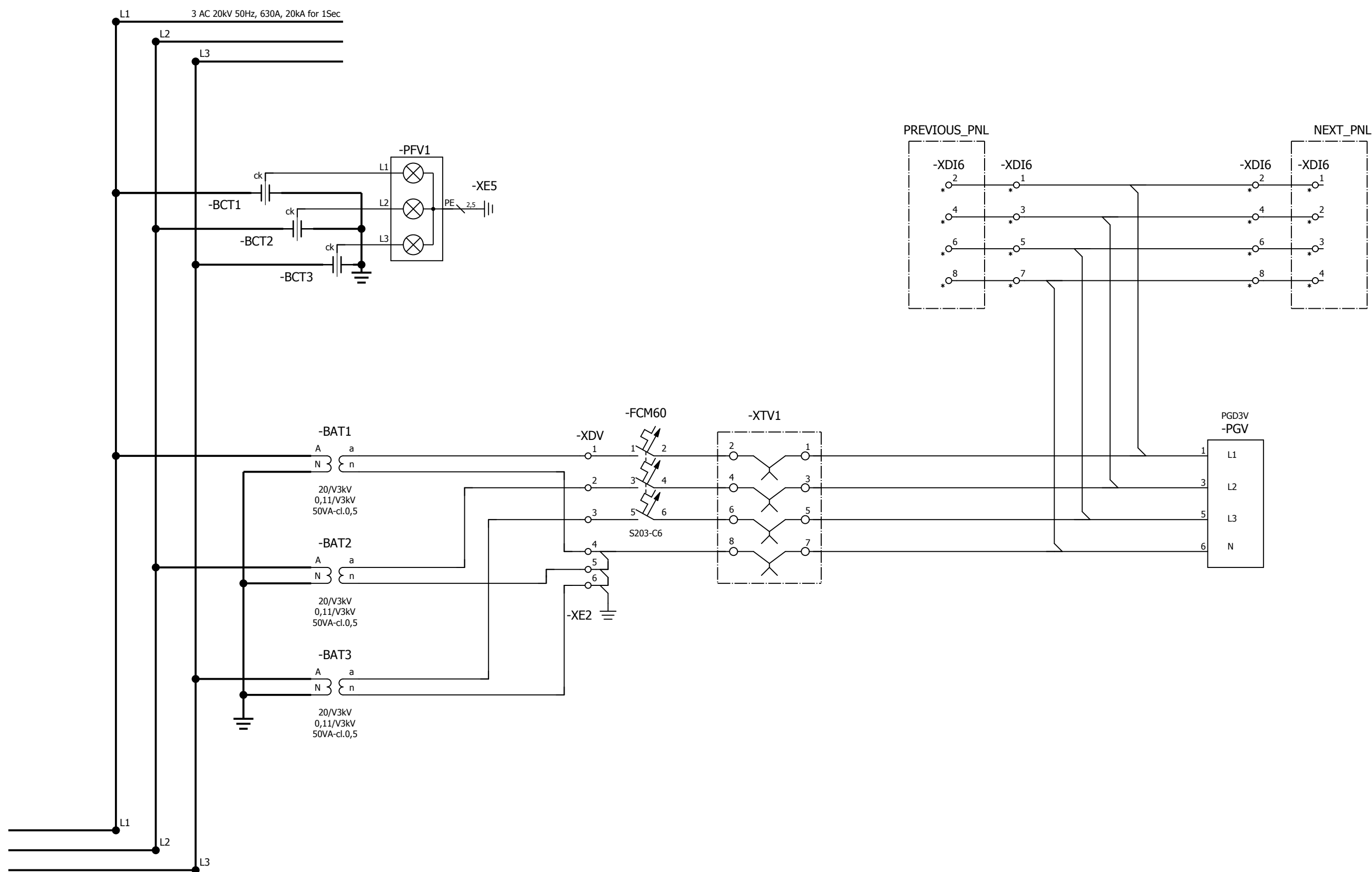
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|------|---------------------|----------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------|------------|-----------------------------------------------------|----------|------------------------|------|-------------------------------------------------------------|--|--------------------|---------------------------|-------|----|
| A    | -TFA                | ACTIVE POWER TRANSDUCERS                                                                           | -XD11 | TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS                                       | A          |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -TFC                | CURRENT TRANSDUCERS                                                                                | -XD12 | TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
| B    | -TFF                | FREQUENCY TRANSDUCERS                                                                              | -XD13 | TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION    | B          |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -TFJ                | ACTIVE ENERGY TRANSDUCERS                                                                          | -XD14 | TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS                                |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -TFK                | REACTIVE ENERGY TRANSDUCERS                                                                        | -XD16 | TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS            |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -TFM                | MULTIFUNCTION TRANSDUCERS                                                                          |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -TFP                | POWER-FACTOR TRANSDUCERS                                                                           | -XD17 | TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS                                                 |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -TFQ                | REACTIVE POWER TRANSDUCERS                                                                         | -XD18 | TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS                                       |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -TFS                | SIGNAL CONVERTERS                                                                                  | -XD19 | TERMINAL BLOCK FOR SPECIAL USE                                                                |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -TFV                | VOLTAGE TRANSDUCERS                                                                                | -XDJ  | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD                                   |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -WA                 | BUSBARS                                                                                            | -XDJ1 | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS                             |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -WBC                | POWER CABLES                                                                                       | -XDM  | SEALABLE TERMINAL BLOCK FOR MEASUREMENT                                                       |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
| C    | -WE                 | EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.                                                   | -XDS  | CURRENT SOCKETS                                                                               | C          |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -WGC                | CONTROL CABLES                                                                                     | -XDT  | TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB                           |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -WGS                | COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)                                                      | -XDU  | CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK                        |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -WH                 | OPTICAL FIBRES                                                                                     | -XDU1 | TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK                                |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDA                | TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS                                                | -XDV  | TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS                                           |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDA1               | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1                                 | -XDV1 | CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1                            |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDA2               | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2                                 | -XDV2 | CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2                            |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDA3               | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3                                 | -XDV3 | CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3                            |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDA5               | CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER                                   | -XDV4 | CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR                                         |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDB                | CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS | -XDX  | SUPPORT TERMINAL BLOCK                                                                        |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
| D    | -XDB1               | CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)           | -XE1  | EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE                                        | D          |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDB2               | CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER                                     | -XE2  | EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE                                       |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDB10...-XDB89     | CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS                                                    | -XE5  | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE                           |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDB7               | CONNECTOR FOR TRUCK POSITION CONTACTS                                                              | -XE6  | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE                                 |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDB91              | CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR                                      | -XE7  | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE                                        |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDB92              | CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR                                      | -XE8  | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE                                |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDB93              | CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR                                             | -XE9  | EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT                                |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDB95              | CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET                                           |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDC                | CUSTOMER TERMINAL BLOCK                                                                            |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDC1               | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER                                  |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
| E    | -XDC2               | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR                              |       |                                                                                               | E          |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDC3               | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT          |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDC4               | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS                             |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDC5               | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER                        |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDE                | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE                                      |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDE1               | TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE                                       |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDE2               | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2                            |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDE3               | TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2                             |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDF                | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS                                |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDF1               | CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR                                            |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
| F    | -XDF2               | CONNECTOR FOR REAR VENTILATOR CIRCUITS                                                             |       |                                                                                               | F          |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDF3               | CONNECTOR FOR FRONT VENTILATOR CIRCUITS                                                            |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDH                | TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)   |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
|      | -XDI                | TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)                                     |       |                                                                                               |            |                                                     |          |                        |      |                                                             |  |                    |                           |       |    |
| 1    |                     | 2                                                                                                  |       | 3                                                                                             |            | 4                                                   |          | 5                      |      | 6                                                           |  | 7                  |                           | 8     |    |
|      |                     |                                                                                                    |       | Date Prep.                                                                                    | 15.11.2018 | PROJECT TITLE<br>MV Switchgear for MNA Serang Plant |          | ABB                    | EPMV | REFERENCE DESIGNATIONS<br>MVD-20 & MVD-21 20kV Boiler Plant |  | Scale              |                           | = H06 |    |
|      |                     |                                                                                                    |       | Prepared                                                                                      | YR         |                                                     |          |                        |      |                                                             |  | Customer           | PT Multimas Nabati Asahan | & 310 |    |
| 00   | Issued for Approval | 11.01.19                                                                                           | YR    | Checked                                                                                       | AW         |                                                     |          | PT. ABB Sakti Industri |      | SCHEMATIC DIAGRAM<br>SDS RISER WITH VT                      |  | Drawing Nbr.       | IDUNISEC-BE318042-301     | Sh.   | 6  |
| Ind. | Revision            | Date                                                                                               | Name  | Approved                                                                                      | AI         | Based on:                                           | BE318042 |                        |      |                                                             |  | Customer Doc. Nbr. |                           | Cont. | 16 |
| 1    |                     | 2                                                                                                  |       | 3                                                                                             |            | 4                                                   |          | 5                      |      | 6                                                           |  | 7                  |                           | 8     |    |

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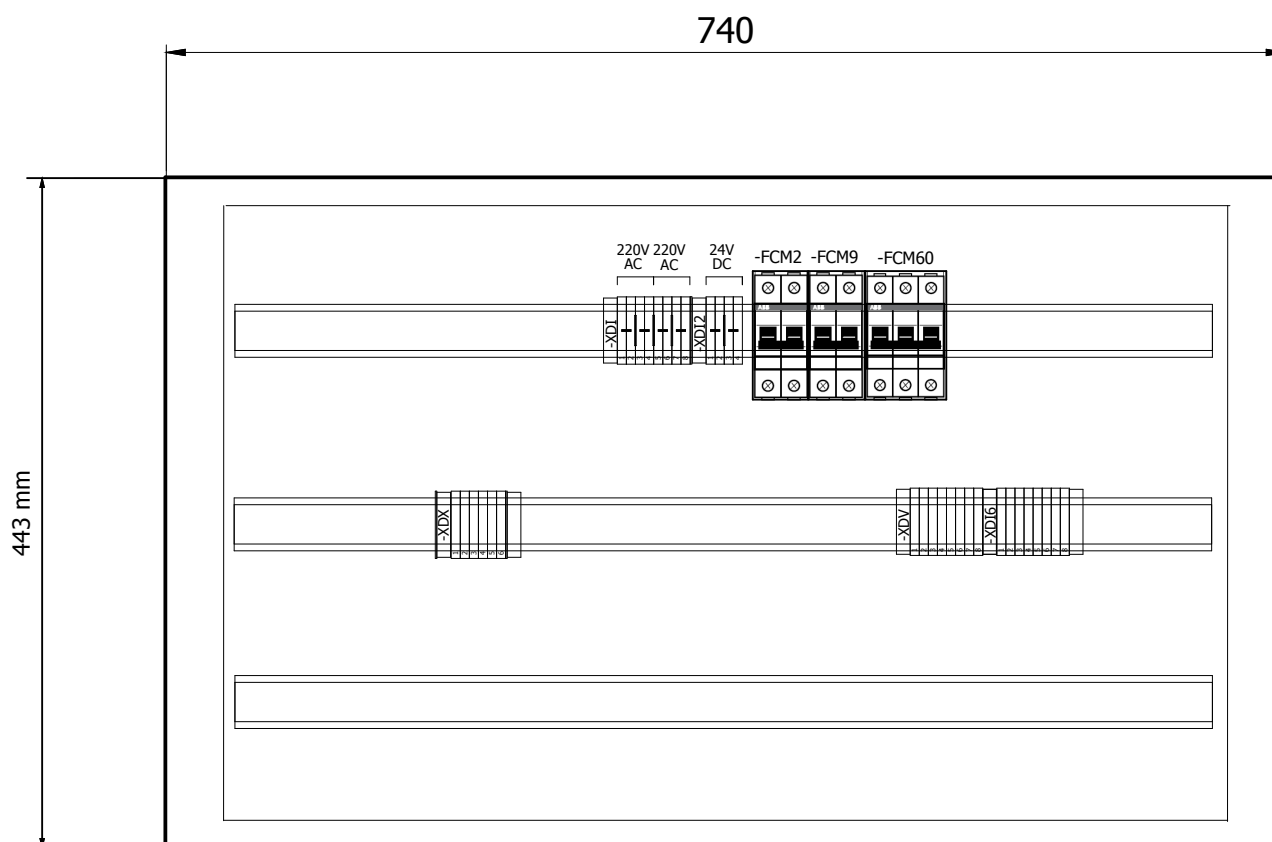


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|      |                     |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br><br>MV Switchgear for MNA Serang Plant | ABB      | EPMV               | MAIN CIRCUITS<br>MVD-20 & MVD-21 20kV Boiler Plant<br>SCHEMATIC DIAGRAM<br>SDS RISER WITH VT | Scale        |                           | = H06 |
|      |                     |          |      | Prepared   | YR         |                                                         |          |                    |                                                                                              | Customer     | PT Multimas Nabati Asahan | & 310 |
| 00   | Issued for Approval | 11.01.19 | YR   | Checked    | AW         |                                                         |          |                    |                                                                                              | Drawing Nbr. | IDUNISEC-BE318042-301     | Sh. 8 |
| Ind. | Revision            | Date     | Name | Approved   | AI         | Based on:                                               | BE318042 | Customer Doc. Nbr. |                                                                                              |              | Cont. 16                  |       |
|      |                     |          |      |            |            | PT. ABB Sakti Industri                                  |          |                    |                                                                                              |              |                           |       |

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|      |                     |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br><br>MV Switchgear for MNA Serang Plant | ABB      | EPMV               | LV BOX LAYOUT<br>MVD-20 & MVD-21 20kV Boiler Plant<br>SCHEMATIC DIAGRAM<br>SDS RISER WITH VT | Scale                  |              | = H06                     |        |
|      |                     |          |      | Prepared   | YR         |                                                         |          |                    |                                                                                              | PT. ABB Sakti Industri | Customer     | PT Multimas Nabati Asahan | & 310  |
| 00   | Issued for Approval | 11.01.19 | YR   | Checked    | AW         |                                                         |          |                    |                                                                                              |                        | Drawing Nbr. | IDUNISEC-BE318042-301     | Sh. 10 |
| Ind. | Revision            | Date     | Name | Approved   | AI         | Based on:                                               | BE318042 | Customer Doc. Nbr. |                                                                                              |                        |              | Cont. 16                  |        |

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|-----------------------|--|----------------------------------------------------------------------------|--|--|--|------------------|--|-----------------|--|----------|---------|---------------------|--|
| Summarized parts list |  |                                                                            |  |  |  |                  |  |                 |  |          |         | IDABB_F02_005-IDABB |  |
|                       |  |                                                                            |  |  |  |                  |  |                 |  |          |         | Carry over: 0.00    |  |
| TAG ID                |  | DESCRIPTION                                                                |  |  |  | MANUFACTUR/BRAND |  | PART NUMBER     |  | QUANTITY | REMARKS |                     |  |
| -EB                   |  | HEATER<br>230VAC, 50W                                                      |  |  |  | SANYANG          |  | CDXN010033P0009 |  | 1        | /7.7:E  |                     |  |
| -FCM2                 |  | MCB S202-C2<br>MCB. 2P, 2A                                                 |  |  |  | ABB              |  | 2CDS252001R0024 |  | 1        | /7.2:C  |                     |  |
| -FCM9                 |  | MCB S202-C6<br>MCB. 2P, 6A                                                 |  |  |  | ABB              |  | 2CDS252001R0064 |  | 1        | /7.6:C  |                     |  |
| -FCM60                |  | MCB S203-C6<br>MCB. 3P, 6A                                                 |  |  |  | ABB              |  | 2CDS253001R0064 |  | 1        | /8.4:C  |                     |  |
| -PGV                  |  | Digital AC Voltmeter<br>Model : DPM PGD3V, 48 X 96 MM                      |  |  |  | RISHABH          |  | DPM PGD3V       |  | 1        | /7.2:D  |                     |  |
| -XTV1                 |  | Voltage Test Block<br>Rated Current : 10A, 250V; Mounting : Flush Mounting |  |  |  | YONGSUNG         |  | YSPTT-04C       |  | 1        | /8.5:C  |                     |  |
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[illegible]